NAME: BOTSHELO MOILWENYANE
STUDENT ID: 2805001389
COURSE: DATA ANALYTICS WITH SQL
EXERCISE 02: SQL AGGREGATE FUNCTIONS & OPERATORS

The students table: A list of students enrolled at the academy, with their age and department.

Student_id	name	age	department
1	Alice	20	IT
2	Bob	22	HR
3	Charlie	21	IT
4	Diana	23	Finance
5	Eve	22	HR

1. List all distinct departments in the students table.

```
SELECT DISTINCT departments
FROM students;
```

department
IT
HR
Finance

Give the average age of students per department

```
SELECT department, AVG (age) AS AVG_age
FROM students
GROUP BY department ;
```

department	AVG_age
IT	20
HR	22
Finance	23

3. Show departments with more than 1 student.

```
SELECT department, COUNT (*) AS student_count
FROM students
GROUP BY department
HAVING COUNT (*) > 1;
```

department	student_count
IT	2
HR	2

4. Get all students whose age is between 21 and 23.

```
SELECT
FROM students
WHERE age BETWEEN 21 AND 23;
```

student_id	name	age	department
2	Bob	22	HR
3	Charlie	21	IT
4	Diana	23	Finance
5	Eve	22	HR

5. List all students in the IT or HR department who are older than 21.

```
SELECT *
FROM students
WHERE department IN ('HR', 'IT')
AND age > 21;
```

student_id	name	age	department
2	Bob	22	HR
5	Eve	22	HR

The courses Table: A catalogue of the courses offered by each department along with the credits awarded.

course_id	course_name	department	credits
101	SQL Basics	IT	3
102	Python	IT	4
103	Data Science	IT	4
104	Excel	Finance	2
105	Statistics	HR	3

Show total credits per department, only for departments with more than 5 total credits.

```
SELECT SUM (credits) AS total credits
FROM courses
GROUP BY department
HAVING SUM (credits) > 5;
```

department	total credits
IT	11

7. List all courses that do not have 4 credits.

```
SELECT *
FROM courses
WHERE course ID NOT IN (4);
```

course_id	course.name	department	credits
101	SQL Basics	IT	3
104	Excel	finance	2
105	Statistics	HR	3

8. Show the to 3 courses by credits in descending order

```
SELECT course_id, course_name, credits
FROM courses
ORDER BY credits DESC
LIMIT 3;
```

course_id	course_name	credits
102	Python	4
103	Data science	4
101	SQL Basics	3

The enrollments Table: A record of which students are enrolled in which courses, with grade each student received.

enrollment_id	student_id	course_id	grade
1	1	101	85
2	2	102	78
3	3	103	90
4	4	104	88
5	5	105	82

1. Give the maximum, minimum, and average grade across all enrollments.

maxi_grade	min_grade	avg_grade
90	78	84.6

```
SELECT
MAX(grade) AS max_grade,
MIN (grade) AS min_grade,
AVG(grade) AS avg_grade,
FROM enrollments;
```

10. Count how many enrollments exist per course.

```
SELECT
    course_id,
COUNT (*) AS enrollment_count
FROM enrollments
GROUP BY course_id ;
```

course_id	enrollment count
101	1
102	1
103	1
104	1
105	1

The salaries Table: A simple payroll record showing salary and bonus amounts per employee, organised by department.

11. Find total salary and total bonus per department

employee id	name	department	salary	bonus
1	Tom	IT	60000	5000
2	Jerry	HR	55000	4000
3	Spike	Finance	70000	6000
4	Tyke	IT	62000	5500
5	Butch	HR	54000	3500

11. Find total salary and total bonus per department

```
SELECT
    department,
SUM (salary) AS total_salary,
SUM (bonus) AS total_bonus,
FROM salaries
GROUP BY Department ;
```

department	total_salary	total_bonus
IT	122 000	10 500
HR	109 000	7 500
Finance	70 000	6 000

12. Show departments where average salary is above 55000.

```
SELECT
        department,
AVG (salary) AS avg_salary
FROM salaries
GROUP BY department
HAVING AVG(salary) > 55000;
```

department	avg_salary
IT	61 000
Finance	70 000

13. List employees whose salary plus bonus is greater than 60000.

```
SELECT
        employee_id,
        name,
        department,
        salary,
        bonus,
FROM salaries
WHERE salary + bonus > 60000;
```

employee_id	name	salary	bonus	total_compensation
1	TOM	60000	5000	65000
4	tyke	62000	5500	67500

The Projects Table: The list of active projects in the ~~counting~~ company, the department running each over, and the budget allocated.

Project_id	Project_name	department	budget
1	AI APP	IT	120000
2	Payroll system	Finance	80000
3	Dashboard	IT	150000
4	Website	Marketing	60000
5	HR portal	HR	50000

14. Show total and average budget per department. Only include departments with average budget above 70000.

```
SELECT
    department,
    Sum(budget) As total_budget,
    AVG (budget) As Avg_budget,
    FROM projects
    GROUP BY department
    HAVING AVG(Budget) > 70000;
```

department	total_budget	avg_budget
IT	270 000	135 000
Finance	80 000	80 000

15. List all Projects with budgets between 50000 and 120000, excluding the Marketing department

```
SELECT
    Project_id,
    Project_name,
    department,
    budget,
FROM Projects
WHERE Budget BETWEEN 50000 AND 120000
AND department <> 'marketing';
```

Project_id	Project name	department	budget
1	AI APP	IT	120000
2	Payroll System	Finance	80 000
5	HR Portal	HR	50000